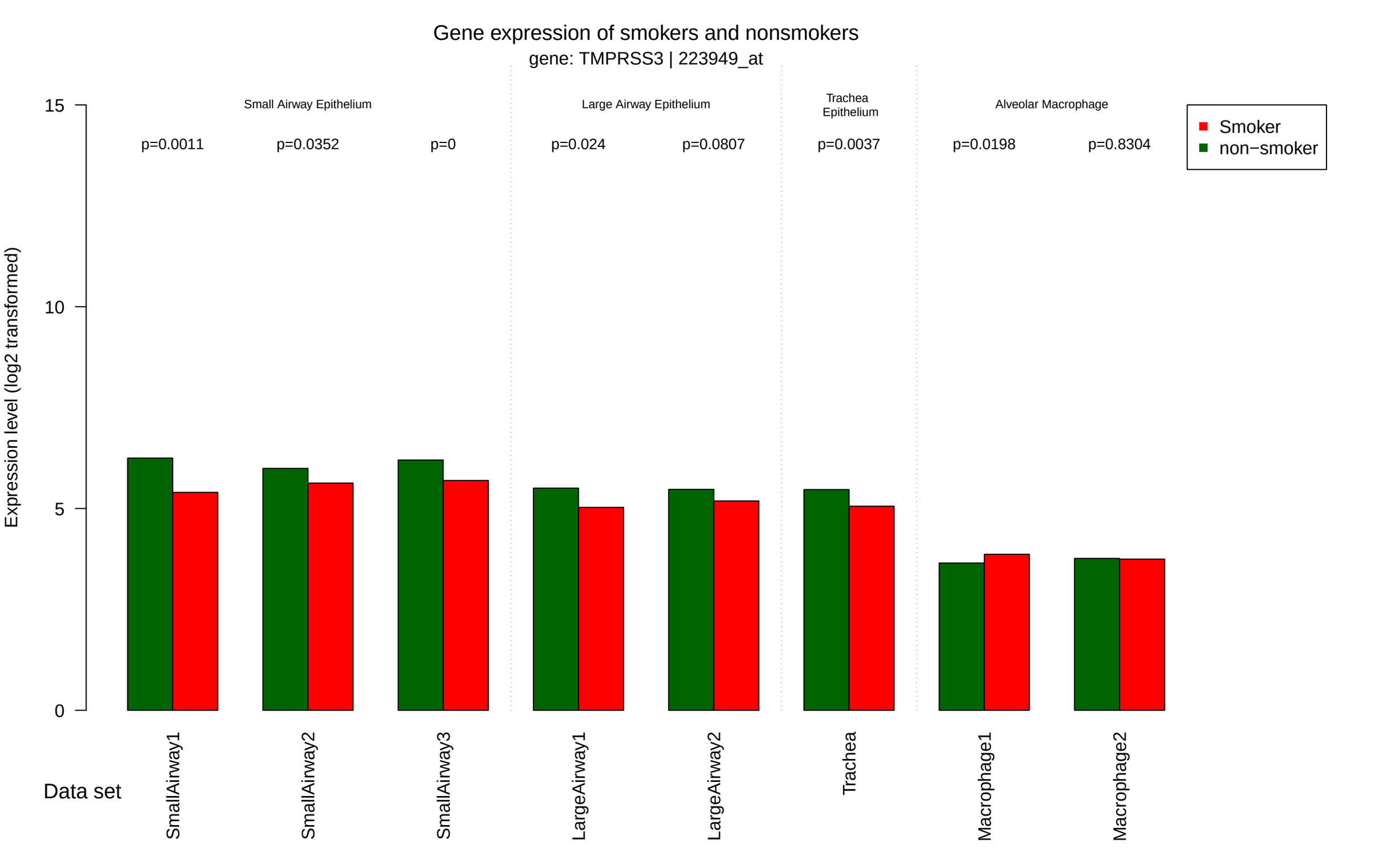
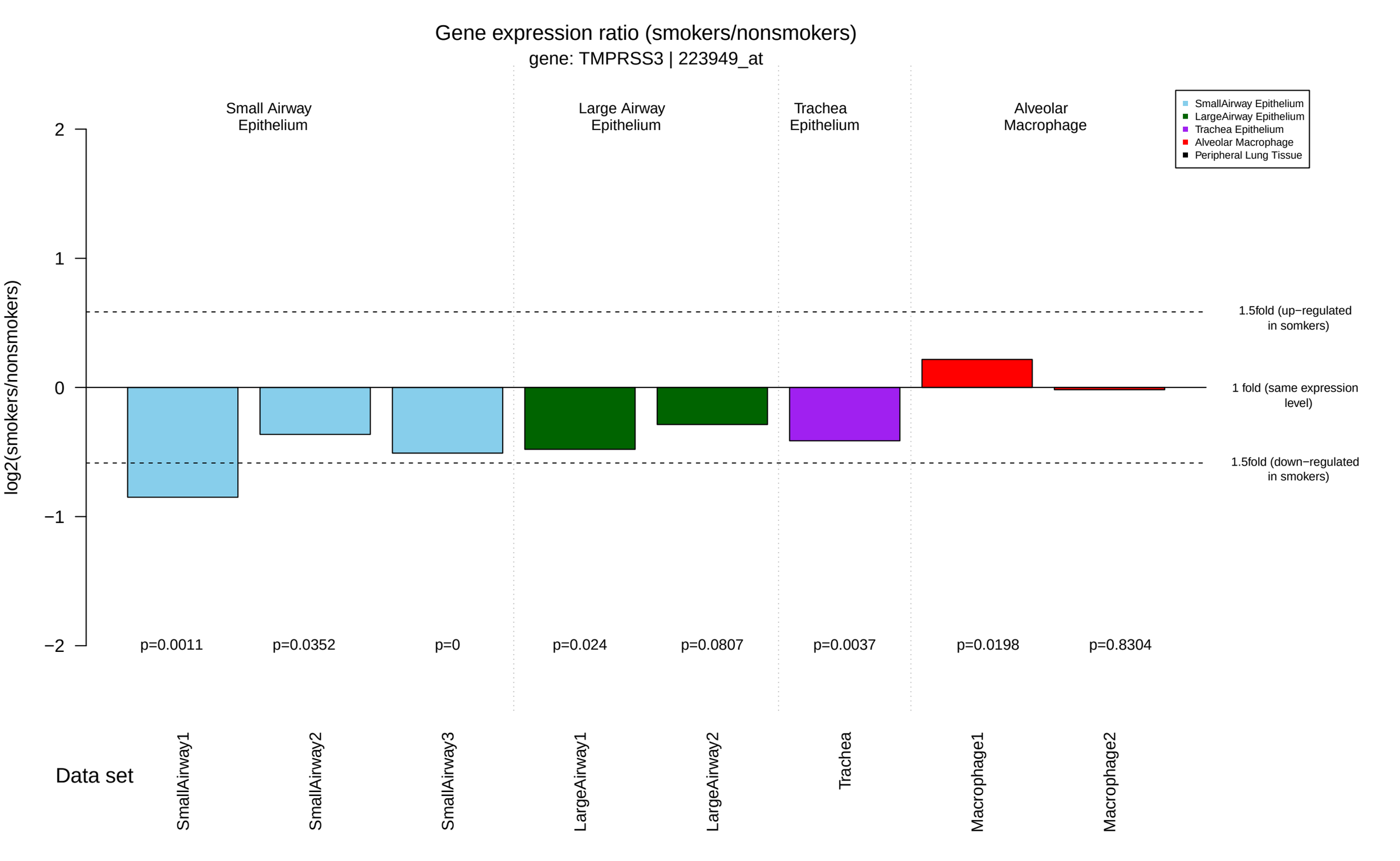
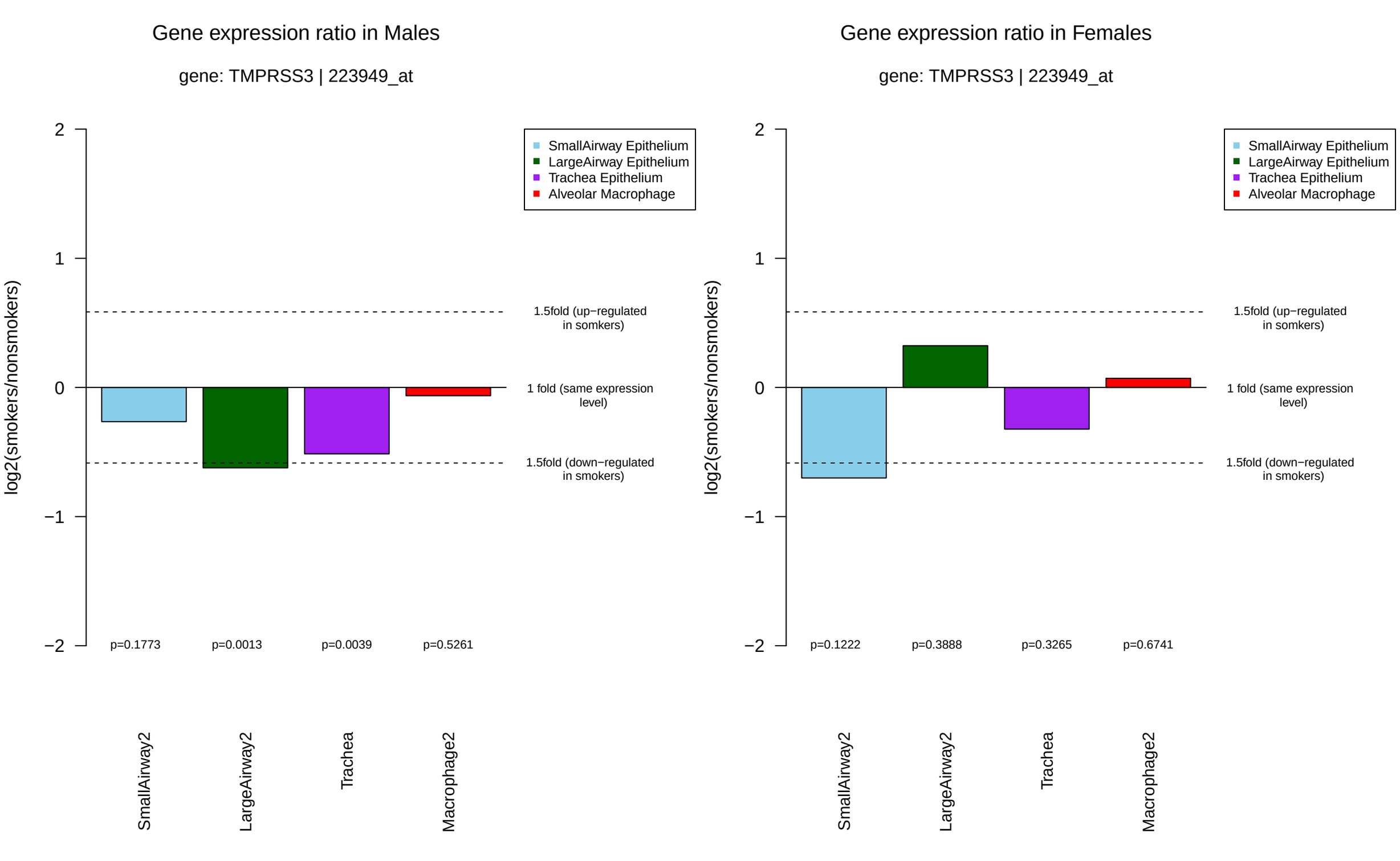
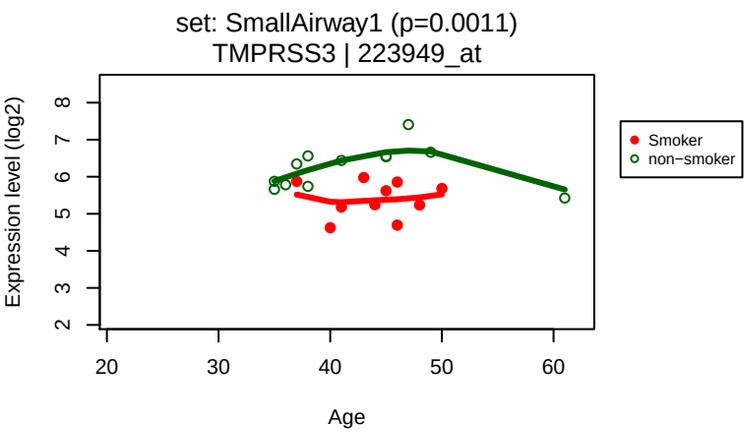
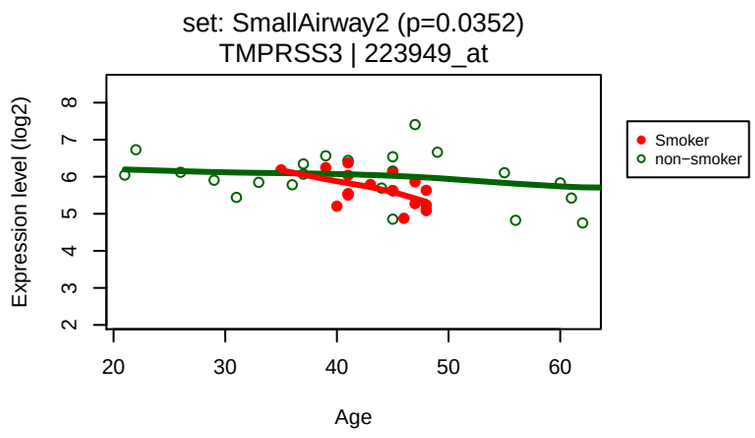
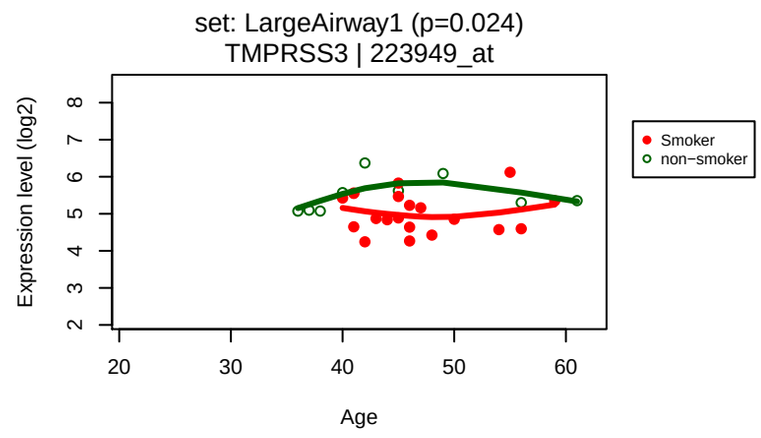

Gene expression of smokers and nonsmokers
gene: TMPRSS3 | 223949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0011
p=0.0352
p=0
p=0.024
p=0.0807
p=0.0037
p=0.0198
p=0.8304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

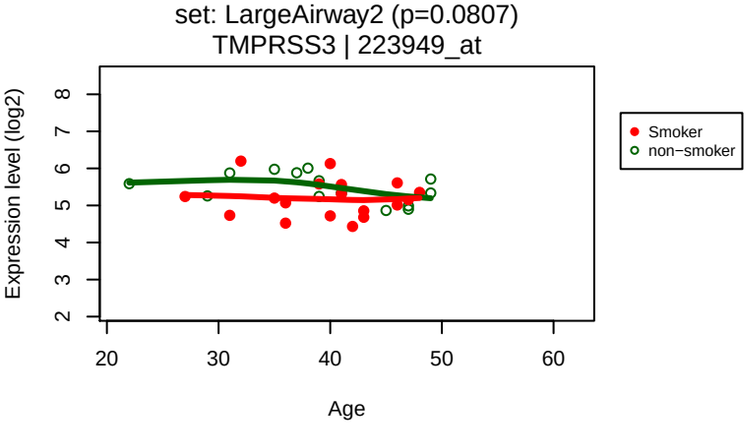

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS3 | 223949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.0352
p=0
p=0.024
p=0.0807
p=0.0037
p=0.0198
p=0.8304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

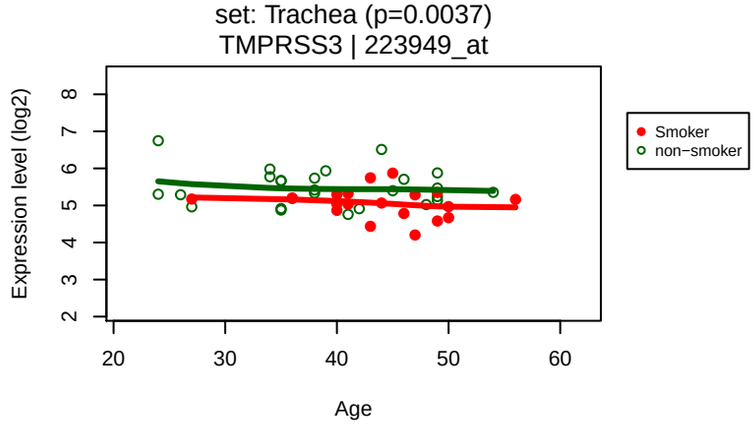

Gene expression ratio in Males
gene: TMPRSS3 | 223949_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1773
p=0.0013
p=0.0039
p=0.5261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS3 | 223949_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1222
p=0.3888
p=0.3265
p=0.6741
SmallAirway2
LargeAirway2
Trachea
Macrophage2

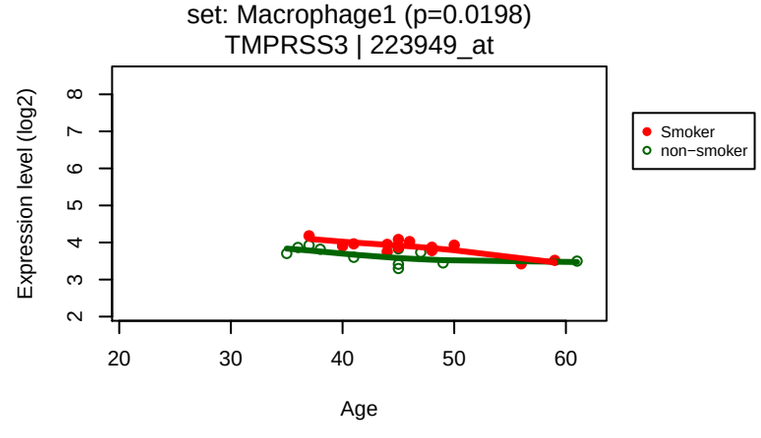

set: SmallAirway1 (p=0.0011)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker

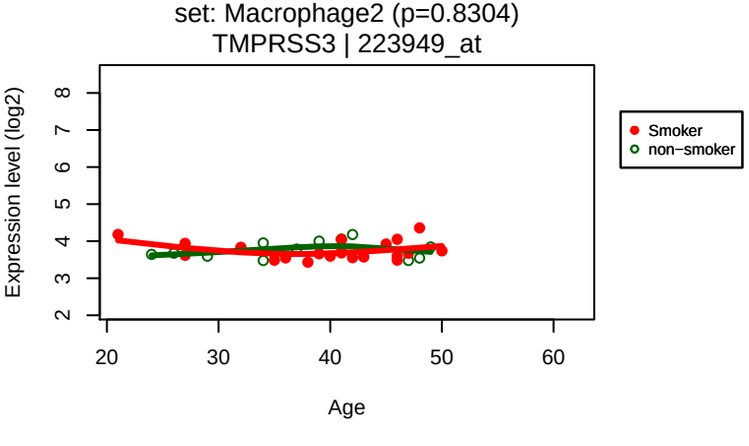

set: SmallAirway2 (p=0.0352)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker

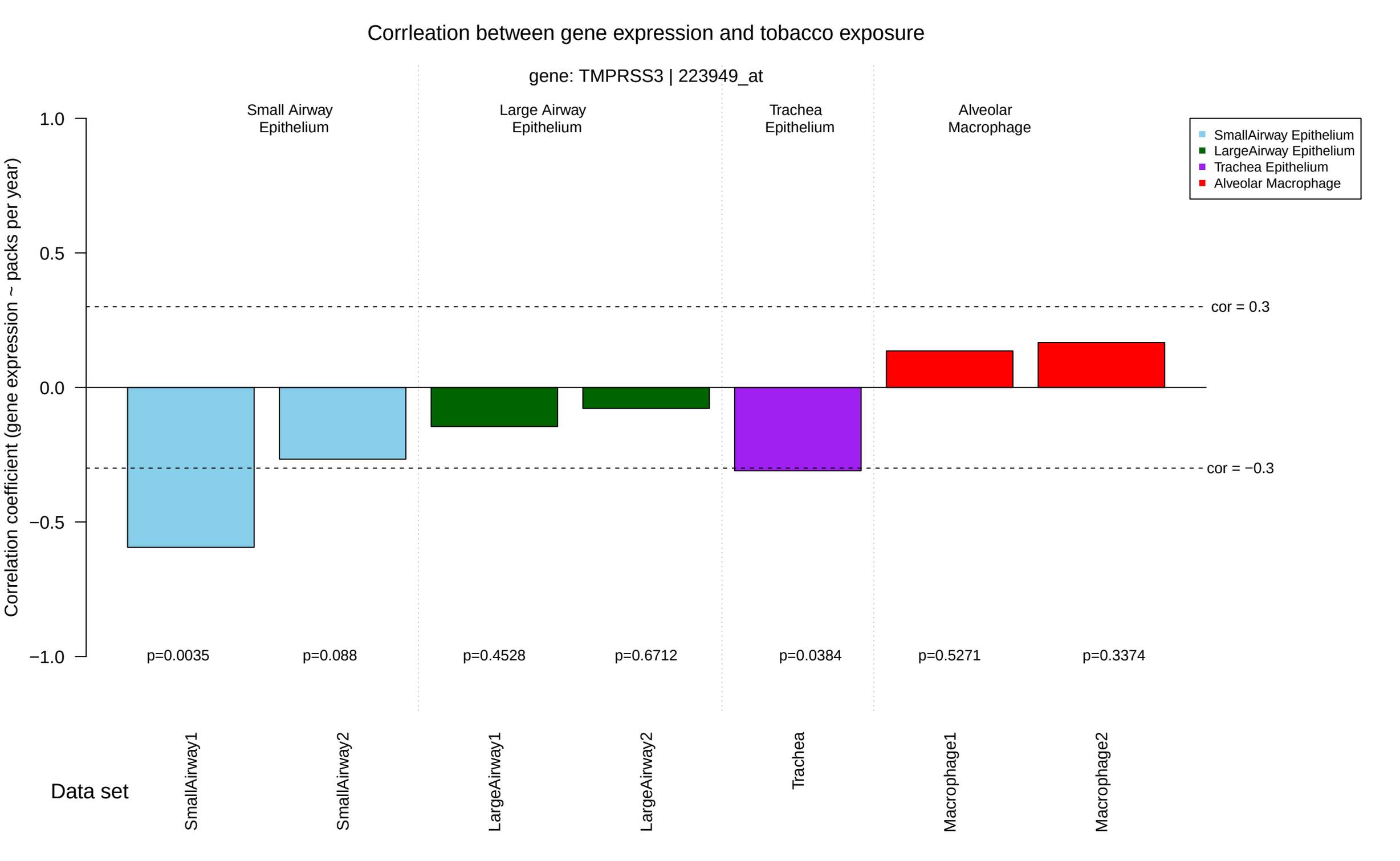

set: LargeAirway1 (p=0.024)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0807)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0198)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8304)
TMPRSS3 | 223949_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS3 | 223949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0035
p=0.088
p=0.4528
p=0.6712
p=0.0384
p=0.5271
p=0.3374
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2